The Boltzmann distribution

$$\frac{p_{liq}}{p_{sol}} = e^{-\frac{\Delta E}{kT}} \; ; \Delta E = -\Delta H \frac{T - T_{mp}}{T_{mp}} \; ; \Delta E = E_{sol} - E_{liq} \Rightarrow p_{liq} = p_{sol}\, e^{\frac{(T-T_{mp})\Delta H}{kTT_{mp}}} \; ; T \approx T_{mp} \Rightarrow p_{liq} = p_{sol}\, e^{\frac{(T-T_{mp})\Delta H}{kT_{mp}^2}} ;$$

$$p_{liq} + p_{sol} = 1 \Rightarrow p_{sol} = 1 - p_{liq} \Rightarrow p_{liq} = \frac{e^{\frac{(T-T_{mp})\Delta H}{kT_{mp}^2}}}{1 + e^{\frac{(T-T_{mp})\Delta H}{kT_{mp}^2}}} \; ; \text{ let's make } b = \frac{kT_{mp}^2}{\Delta H} \Rightarrow$$

$$\Rightarrow \quad p_{liq} = \frac{e^{\frac{(T-T_{mp})}{b}}}{1 + e^{\frac{(T-T_{mp})}{b}}}$$